DICOM Conformance Statement

CardiacStudy Player — Cath, Echo & CT DICOM Viewer for macOS Version 1.1.0 · Document date 2026-11-06 · Document version 1.2

> **Disclaimer.** CardiacStudy Player is intended for review, teaching and educational use only. It is **not a certified medical device** and is not intended for primary diagnosis or clinical decision-making. This document is structured after DICOM PS3.2 but has not been independently audited.

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1. Conformance Statement Overview

CardiacStudy Player is a standalone macOS application that reads DICOM PS3.10 (Part 10) files and DICOMDIR file-sets from local storage or mounted removable media (CD/DVD/USB), and displays X-ray angiographic (XA) cine runs, multiframe/single-frame ultrasound (US) images, and the Encapsulated PDF and Structured Report (SR) documents contained in a study. It can export loaded runs to QuickTime (.mov) movies and copy the displayed frame to the system clipboard (both non-DICOM outputs).

CardiacStudy Player implements **Media Interchange services only**, in the **File-Set Reader (FSR)** role. It provides **no network services** of any kind.

Table 1-1. Network Services

Service	User of Service (SCU)	Provider of Service (SCP)
All DIMSE services (Verification, Storage, Query/Retrieve, Print, Storage Commitment, Worklist, MPPS)	No	No
DICOMweb (STOW-RS / QIDO-RS / WADO-RS)	No	No

The application never opens a network association or socket for DICOM purposes; it runs fully offline.

Table 1-2. Media Services

Media Storage Application Profile	Write Files (FSC/ FSU)	Read Files (FSR)
General Purpose CD-R / DVD (STD-GEN-CD, STD-GEN-DVD-JPEG)	No	Yes
Basic Cardiac X-Ray Angiographic (STD-XABC-CD)	No	Yes
1024 X-Ray Angiographic (STD-XA1K-CD, STD-XA1K-DVD)	No	Yes
Ultrasound (STD-US-ID/SC/CC-MF…, CD and DVD media)	No	Yes

CT/MR (STD-CTMR-… CD and DVD media)	No	Yes

CardiacStudy Player never creates or updates a DICOM file-set and never writes DICOM objects.

2. Introduction

2.1 Revision History

Doc version	Date	Application version	Notes
1.2	2026-11-06	1.1.0	Slice position, reference view and slice marks for CT/MR series (§4.6)
1.1	2026-10-23	1.0.0	Written against DICOM 2026d (§2.2); adds the JPEG XL, Deflated Image Frame Compression and Encapsulated Uncompressed transfer syntaxes (§4.5)
1.0	2026-10-19	1.0.0	Initial release

2.2 Audience and Remarks

This document is intended for people integrating or evaluating CardiacStudy Player alongside other DICOM applications (modalities, media creators, PACS export functions). It assumes familiarity with the DICOM standard (PS3.1–PS3.20).

This statement is written against the **2026d edition** of the DICOM Standard, the current edition at its date. The SOP Classes (§4.4) and transfer syntaxes (§4.5) are as that edition defines them, including the JPEG XL and Deflated Image Frame Compression transfer syntaxes added since 2024. DICOM editions are backward compatible for the objects this application reads, so files from equipment built to earlier editions are read the same way.

Conformance to this statement does not guarantee interoperability; validation with representative media from the intended source systems is recommended.

2.3 Definitions and Abbreviations

FSR — File-Set Reader. TS — Transfer Syntax. SOP — Service-Object Pair. VOI — Value of Interest. XA — X-Ray Angiography. US — Ultrasound.

2.4 Implementation Identification

Application name	CardiacStudy Player
Version	1.1.0
Platform	macOS 15+ (universal: native on Apple silicon and Intel)

DICOM toolkit	DICOMKit, the application's own Swift parser and pixel pipeline (native, RLE, JPEG Lossless, JPEG-LS and 8/12-bit JPEG Extended decoders); OpenJPEG 2.5.3 (JPEG 2000, HTJ2K); Apple ImageIO (8-bit JPEG Baseline, JPEG XL); Apple AVFoundation (video transfer syntaxes); Apple PDFKit (Encapsulated PDF)

3. Networking

Not applicable. CardiacStudy Player implements no DICOM network application entities, initiates no associations, and accepts none.

4. Media Interchange

4.1 Implementation Model

One local Application Entity, **MEDIA-FSR**, is activated by the user through the GUI (File ▸ Open DICOM File…/Open Study Folder…, or the equivalent buttons). It reads a file-set or individual files, builds an index of the image and document instances found (metadata only), and then decodes the pixel data of a run on demand when the user selects it. Decoded frames are held in memory only; nothing is written back to the media, which can be ejected from within the application.

4.2 Real-World Activities

- **RWA 1 — Load a file-set:** the user selects a disc/folder. The application locates a `DICOMDIR` at the selected location or one subfolder level below it (case-insensitive name match) and enumerates every existing file referenced by Referenced File ID (0004,1500). Directory-record hierarchy attributes are not otherwise interpreted. When the user opens a `DICOMDIR` *file* directly, the App Sandbox (§6) first asks the user to grant access to its folder, so that the files it references can be read.
- **RWA 2 — Load individual files:** the user selects a single Part 10 file, or a folder containing `*.dcm` files (searched recursively) without a DICOMDIR.
- **RWA 3 — Review:** indexed instances appear as thumbnails; the selected instance is displayed with cine playback, frame stepping, window/level, zoom, and edge-enhancement controls. Encapsulated PDF instances are rendered as pageable images (or opened in an external PDF viewer), and Structured Report instances as pageable text renderings of their content tree.

4.3 Media Application Profiles — Roles

CardiacStudy Player is a **general-purpose FSR**: it does not verify that a file-set declares or conforms to a specific application profile. Any file-set whose instances use the SOP classes and transfer syntaxes listed below can be read; this covers the STD-GEN, STD-XABC, STD-XA1K, STD-US, and STD-CTMR family profiles listed in Table 1-2. CT and MR studies are also commonly issued under IHE **Portable Data for Imaging (PDI)** rather than a PS3.11 STD- profile; PDI media build on the general-purpose CD/

DVD profile and are read the same way. Because CardiacStudy Player never validates a profile, this list describes the media it is designed against, not a gate applied to the file-set.

4.4 SOP Classes

Instance selection is **content-based, not SOP-Class-restricted**: any instance whose pixel data can be decoded is displayed. The SOP classes below are the design targets and the ones exercised in testing.

SOP Class	UID	Support
X-Ray Angiographic Image Storage	1.2.840.10008.5.1.4.1.1.12.1	Display, cine
Enhanced XA Image Storage	1.2.840.10008.5.1.4.1.1.12.1.1	Display, cine (frame timing from per-frame Frame Content macros)
X-Ray Radiofluoroscopic Image Storage (RF)	1.2.840.10008.5.1.4.1.1.12.2	Display, cine — same pipeline as XA (content-based selection; not a distinct code path)
CT Image Storage	1.2.840.10008.5.1.4.1.1.2	Single-frame slices sharing a Series Instance UID are grouped into one browsable multi-slice run, ordered by through-plane geometry (§4.6), split into one run per reconstruction or cardiac phase where the series carries more than one, and stepped (Left/Right or the frame slider) — not cine-played, since the "frames" are separate files with no playback timing of their own (§4.6)
MR Image Storage	1.2.840.10008.5.1.4.1.1.4	Single-frame slices are grouped, ordered, stepped and window-levelled exactly as CT above — the grouping is modality-agnostic. A series carrying more than one echo is split into one run per echo, identified by Echo Numbers (0018,0086) where present, else Echo Time (0018,0081) (§4.6)
Enhanced MR Image Storage	1.2.840.10008.5.1.4.1.1.4.1	Display, multi-frame stepping through the instance's own frames; per-frame functional groups are read for windowing as for other enhanced objects. Not further partitioned — a single instance is already one run
Ultrasound Image Storage	1.2.840.10008.5.1.4.1.1.6.1	Display (single frame)
Ultrasound Multi-frame Image Storage	1.2.840.10008.5.1.4.1.1.3.1	Display, cine
Encapsulated PDF Storage	1.2.840.10008.5.1.4.1.1.104.1	Rendered per page (also detected by an `application/pdf` Encapsulated Document when the SOP Class UID is absent)
SR Document Storage family (Basic Text, Enhanced, Comprehensive, …)	1.2.840.10008.5.1.4.1.1.88.*	Content tree rendered as read-only paginated text (also detected by Modality SR with a Content Sequence when the SOP Class UID is absent). TEXT/CODE/NUM/ PNAME/DATE-TIME/UIDREF items shown; IMAGE/ COMPOSITE/WAVEFORM references listed by SOP Instance UID, not resolved

Waveform Storage family (12-lead/General/ Ambulatory ECG, Hemodynamic, Cardiac Electrophysiology, …)	1.2.840.10008.5.1.4.1.1.9.*	Each multiplex group's channels rendered as paginated, calibrated lead-strip images — a channel's trace is in physical units (e.g. µV) when Channel Sensitivity is present, else raw digital values; also detected by the presence of a Waveform Sequence when the SOP Class UID is absent
Secondary Capture and other image SOP classes	various	Best-effort display when pixel data is decodable
Grayscale Softcopy Presentation State Storage	1.2.840.10008.5.1.4.1.1.11.1	Rendered as one read-only page per referenced image: the Presentation State's own Modality LUT (Rescale Slope/Intercept) when present, else the referenced image's; its Softcopy VOI LUT Sequence window, else the referenced image's own; Presentation LUT Shape polarity; a Displayed Area Selection crop; and its Graphic Annotation Sequence (Text Objects, and CIRCLE/ELLIPSE/POLYLINE/INTERPOLATED/POINT Graphic Objects), layered by Graphic Layer Order (§4.6)
XA/XRF Grayscale Softcopy Presentation State Storage	1.2.840.10008.5.1.4.1.1.11.5	Same rendering as Grayscale Softcopy Presentation State Storage above
Media Storage Directory Storage (DICOMDIR)	1.2.840.10008.1.3.10	Read (Referenced File ID resolution only)

Color, Pseudo-Color, Blending, and Volumetric (MPR / Volume Rendering) Presentation States are **not** interpreted as such. A Key Object Selection Document is rendered as paginated text of its content tree, like any other document in the SR SOP Class family, but the selection itself is **not** acted on: its references are listed rather than resolved, and the instances it selects are neither marked nor gathered anywhere in the interface. Any image instance whose transfer syntax is a video codec (including US Multi-frame studies exported as MPEG-4) is decoded and played per 4.5. Structured Report documents are rendered as read-only paginated text of their content tree; coded entries are shown by their Code Meaning (coding scheme designators and values are not displayed), and templates are not validated. IMAGE/COMPOSITE references inside an SR content tree are listed by SOP Instance UID rather than resolved to the referenced instance, even when that instance is a Waveform run present elsewhere in the same study.

4.5 Transfer Syntaxes

Decoding is provided by the application's own Swift decoders, the statically linked OpenJPEG library, and the macOS system frameworks ImageIO and AVFoundation (§2.4). Every decoder ships in, or with, the operating system the application requires; nothing is downloaded.

Transfer Syntax	UID	Read
Implicit VR Little Endian	1.2.840.10008.1.2	Yes
Explicit VR Little Endian	1.2.840.10008.1.2.1	Yes

Deflated Explicit VR Little Endian	1.2.840.10008.1.2.1.99	Yes
Explicit VR Big Endian (retired)	1.2.840.10008.1.2.2	Yes
RLE Lossless	1.2.840.10008.1.2.5	Yes
JPEG Baseline (Process 1)	1.2.840.10008.1.2.4.50	Yes
JPEG Extended (Process 2 & 4)	1.2.840.10008.1.2.4.51	Yes, 8- and 12-bit, by the application's own sequential DCT decoder (Huffman); a scan header declaring an invalid spectral range is repaired before decoding (§4.5). Lossy output may differ from libjpeg's by ±1 level, as ITU-T T.81 permits
JPEG Lossless (Process 14)	1.2.840.10008.1.2.4.57	Yes
JPEG Lossless SV1 (Process 14, FOP=1)	1.2.840.10008.1.2.4.70	Yes
JPEG-LS Lossless / Near-Lossless	1.2.840.10008.1.2.4.80 / .81	Yes
JPEG 2000 Lossless / Lossy	1.2.840.10008.1.2.4.90 / .91	Yes
MPEG-4 AVC/H.264 (all profiles)	1.2.840.10008.1.2.4.102–.106	Yes — AVFoundation
HEVC/H.265	1.2.840.10008.1.2.4.107 / .108	Yes — AVFoundation
MPEG2 Main Profile / High Profile	1.2.840.10008.1.2.4.100 / .101	Attempted through AVFoundation; not verified, and depends on the MPEG-2 decoding macOS provides. A run that cannot be decoded shows **Error** (below)
High-Throughput JPEG 2000 (HTJ2K)	1.2.840.10008.1.2.4.201–.203	Yes — OpenJPEG
JPEG XL Lossless	1.2.840.10008.1.2.4.110	Yes — ImageIO, 8- and 16-bit allocated (see below)
JPEG XL JPEG Recompression	1.2.840.10008.1.2.4.111	Yes — ImageIO
JPEG XL	1.2.840.10008.1.2.4.112	Yes — ImageIO, lossy or lossless
Encapsulated Uncompressed Explicit VR Little Endian	1.2.840.10008.1.2.1.98	Yes — each fragment read as one Native-format frame
Deflated Image Frame Compression	1.2.840.10008.1.2.8.1	Yes — each fragment inflated (RFC 1951) and read as one Native-format frame; 8- and 16-bit allocated only (see below)

JPEG XL frames are decoded by the macOS ImageIO framework, one frame per fragment (PS3.5 A.4.12). As PS3.5 8.2.15 requires, the bitstream's own description of the pixels governs decoding: colour

frames are delivered as RGB whatever the Photometric Interpretation says, and samples are taken from the decoder's bitmap without colour matching. Where a bitstream declares fewer bits than it is decoded to (for example 10-bit stored in 16 allocated), the samples are scaled back to the declared depth so stored values match the encoder's input exactly. Bits Allocated 24 is not read, and signed (Pixel Representation 1) JPEG XL data, which the standard leaves to the bitstream, has not been verified against a real encoder.

Encapsulated Uncompressed and Deflated Image Frame Compression frames go through the same Native-format reader as uncompressed files, so Planar Configuration and YBR_FULL_422 are handled as in §4.6. Bits Allocated 1, the main use of Deflated Image Frame Compression (Segmentation objects), is not displayed, as for uncompressed files.

The following transfer syntaxes defined in DICOM 2026d are **not** read: JPEG 2000 Part 2 Multi-component (1.2.840.10008.1.2.4.92/.93); the JPIP referenced-pixel syntaxes (1.2.840.10008.1.2.4.94/.95, .204/.205), which point to a network server rather than carrying pixels on the media; the SMPTE ST 2110 syntaxes (1.2.840.10008.1.2.7.*), which are for real-time streaming; and the retired JPEG processes. An instance in one of these shows **Error**, as below.

Video-transfer-syntax runs are decoded whole into memory as RGB frames (the encapsulated bitstream is reconstructed across fragments), capped at 1000 frames per run; window/level does not apply to them. When the data set carries no frame-rate attribute, the rate reported by the video stream itself is used. The Fragmentable variants (the `.1` suffixes) of the video transfer syntaxes are read the same way. An instance in an unsupported transfer syntax is indexed but its thumbnail shows **Error** and the failure is logged to the in-app Console; it does not block the rest of the study.

One encoder defect is tolerated rather than reported. ITU-T T.81 requires a sequential JPEG scan header to span the full spectral range (Ss=0, Se=63), but some encoders emit sequential frames declaring a DC-only range, which a conforming decoder refuses even though the entropy-coded data is intact. Where a frame's scan header shows this, the declared range is corrected and the frame decoded normally; the repair is logged to the Console. It is applied only to sequential frame headers, never to progressive or lossless ones, whose narrower spectral ranges are legitimate.

4.6 Display Pipeline (Attribute Usage)

- **Photometric Interpretation:** MONOCHROME1 (rendered inverted), MONOCHROME2, PALETTE COLOR (color LUT applied), RGB, YBR_ICT and YBR_RCT (inverse component transform applied by the JPEG 2000 decoder), and YBR_FULL / YBR_FULL_422. Lossy JPEG decoders convert YCbCr to RGB themselves; for every other transfer syntax (uncompressed, RLE Lossless, JPEG Lossless, JPEG-LS) 8-bit YBR_FULL and YBR_FULL_422 data is converted to RGB with the PS3.3 C.7.6.3.1.2 full-range equations, matching pydicom exactly. Uncompressed YBR_FULL_422 is first expanded from its two bytes per pixel (Y1 Y2 Cb Cr per pixel pair).
- **Modality LUT:** a Modality LUT Sequence (0028,3000) is applied to grayscale data when present; otherwise the linear Rescale Slope (0028,1053) / Intercept (0028,1052).
- **VOI:** grayscale images are windowed with the VOI LUT Function (0028,1056) the data set declares — LINEAR, LINEAR_EXACT, or SIGMOID (PS3.3 C.11.2). The default window is resolved from, in order: the first value pair of the top-level Window Center (0028,1050) / Window Width (0028,1051); the Frame VOI LUT functional-group macro (0028,9132) — shared group first,

then the first per-frame item (per-frame *varying* windows are not reproduced); the input range of the first VOI LUT Sequence (0028,3010) item; the full data range. When a VOI LUT Sequence is present its first LUT is rendered as the default view (output normalized over the table's value range); adjusting window/level interactively replaces the table with a window, and Reset returns to the table.

- **Pixel geometry:** non-square pixels are displayed square. The vertical:horizontal ratio is taken from Pixel Aspect Ratio (0028,0034), or from the row:column ratio of Pixel Spacing (0028,0030) / Imager Pixel Spacing (0018,1164) / the Enhanced shared Pixel Measures group; the lower-resolution axis is upscaled (never downsampled). Only the ratio is used here — see Calibrated distance measurement below for the one place an absolute spacing value is used.
- **Lossy compression indication:** the Lossy Image Compression state (0028,2110), with method (0028,2114) and ratio (0028,2112) when present, is shown in the Patient panel for the selected instance.
- **Cine timing:** the playback rate is resolved from, in order: XA Acquisition Frame Rate → Cine Rate → Recommended Display Frame Rate → Frame Time → mean of Frame Time Vector → Actual Frame Duration → for Enhanced multiframe objects, the median inter-frame interval of the per-frame Frame Acquisition/Reference DateTime values. A Frame Time Vector's per-frame variable timing is reproduced during playback: each frame's dwell is its vector entry, scaled proportionally when the user adjusts the rate (whose readout stays the mean rate). Non-positive entries fall back to the mean interval. QuickTime export renders at a single uniform rate.
- **Overlay Planes:** the 60xx Overlay Plane groups (Overlay Data bitmaps) are composited in white onto monochrome images after the VOI stage. A single-frame overlay applies to every frame of a multi-frame image; a multi-frame overlay maps through Image Frame Origin; Overlay Origin places the plane (clipped at the image edges). Overlays embedded in unused Pixel Data bits (retired in DICOM 2004) are not extracted, and overlays are not composited onto color images.
- **Display Shutter:** RECTANGULAR, CIRCULAR, and POLYGONAL shutter shapes (PS3.3 C.7.6.11) are combined and the area outside them is painted with the Shutter Presentation Value (high byte; black when absent) on monochrome images. The Enhanced multi-frame Frame Display Shutter macro is not applied.
- **Positioner angles:** Positioner Primary/Secondary Angle (0018,1510/ 0018,1511) — top-level or the Enhanced XA/XRF Positioner macro (0018,9405) — are shown in the Patient panel in the LAO/RAO · CRA/CAU convention (PS3.3 C.8.7.5). A rotational acquisition's per-frame varying angles are not tracked; the first value stands for the run.
- **Mask subtraction (DSA):** a Mask Subtraction Sequence (0028,6100) item with Mask Operation AVG_SUB enables the View ▸ Subtracted (DSA) toggle: the mask is the mean of the Mask Frame Numbers, subtraction is limited to the Applicable Frame Range (other frames show native), and a run whose Recommended Viewing Mode (0028,1090) is SUB starts subtracted. While subtracted, the window defaults span the subtracted signal; QuickTime export and Copy follow the on-screen state. Mask Sub-pixel Shift (0028,6114) is applied as the mask's initial registration shift (bilinear, edge-replicated), and the user can re-register the mask against patient motion in 1 px steps (⌥-arrows, View ▸ Nudge DSA Mask) or 0.1 px steps (⇧⌥-arrows). Mask Operation TID is **not** applied. As an application feature beyond the data set, the user may designate the currently displayed frame as the mask (View menu), which enables subtraction for multi-frame grayscale

cines carrying no Mask Subtraction Sequence and supersedes a file-provided spec until removed.

- **Multi-slice series grouping (CT/CTA):** single-frame image instances that share a Series Instance UID (0020,000E) — a CT/CTA axial series, which arrives as one file per slice with no cine mechanism of its own — are collapsed at index time into one browsable multi-slice run, ordered by through-plane position — Image Position Patient (0020,0032) projected onto the normal derived from Image Orientation Patient (0020,0037), with the direction aligned to the series' Instance Number trend so a well-formed series orders exactly as Instance Number would. Slice Location (0020,1041), Instance Number (0020,0013), SOP Instance UID and the source path break ties or stand in when the geometry is absent. A Series Instance UID that carries more than one reconstruction (Convolution Kernel (0018,1210), Slice Thickness (0018,0050)), cardiac phase (Nominal Percentage of Cardiac Phase (0020,9241)), or MR echo (Echo Numbers (0018,0086), else Echo Time (0018,0081)) becomes one run per stack rather than one interleaved run. The grouping is modality-agnostic: it applies to any single-frame instances sharing a Series Instance UID, CT and MR alike, and a component simply has no effect on a series that does not carry it. Because these attributes are unreliable in practice, a component partitions a series only when it is present on every instance in that series *and* leaves no single-instance group; the decision is made across the whole series, never per slice, and a run is labelled only by those components uniform across it. The echo is named only where it actually distinguishes runs, since an unsplit single-echo series would otherwise be labelled with a first echo that tells the reader nothing. The runs one series becomes are presented in cardiac-phase then echo order — a dual-echo MR opens echo 1 first whatever order its files were read in — while a reconstruction split, having no natural order, keeps the order read; runs of other series are not reordered. Each slice is decoded from its own file on selection (no combined in-memory volume, no interpolation between slices — the same "decode on demand" pattern used for cine frames). The run steps (Left/Right, the frame slider) but does not cine-play, since there is no real inter-slice playback timing; it keeps window leveling and behaves like a normal image run for Quiz Mode, export, and the Caliper Tool — a true multi-frame cine (Number of Frames > 1) is never grouped this way, even when it shares a Series Instance UID with other runs.
- **Slice position and reference view (CT/MR series):** each slice of a grouped series is labelled with its position along whichever patient axis the slice normal (from Image Orientation Patient (0020,0037)) is closest to, read from Image Position Patient (0020,0032): "S 42.5 mm" or "I 12.0 mm" for axial slices, R/L for sagittal, A/P for coronal. Without usable geometry it falls back to Slice Location (0020,1041) ("Loc 42.5 mm"); no label is shown when neither is present. A series of ten or more slices in which every slice carries both attributes, and whose slices do not all lie at one position (within 0.001 mm), also gets a reference view: a line per slice drawn on an image of the stack. Its picture is a localizer from the same study when one qualifies — Image Type (0008,0008) value 3 LOCALIZER, the same Study Instance UID (0020,000D), the same Frame of Reference UID (0020,0052) when both carry one, a plane tilted from the slices' by more than 60°, and crossed by the middle slice; among several, the least tilted, then frontal before lateral before axial. The line is the intersection of each slice plane with the localizer's plane, clipped to the image. With no qualifying localizer the picture is a mean projection of the stack's own pixels. Neither image is altered or saved. As an application feature beyond the data set, the user may name slices (View ▸ Mark Slice); marks are keyed by SOP Instance UID (0008,0018), kept in the application's settings and never written to any DICOM file.
- **Presentation State rendering (GSPS / XA-XRF GSPS):** at index time, each Presentation State's

Referenced Series/Image Sequence is resolved against every other instance in the same input, producing one read-only page per reference it finds there (unresolved references are skipped and logged, not silently dropped). Each page applies, in order: the Presentation State's own Modality LUT (Rescale Slope/Intercept) when present — needed when the referenced image (often a Secondary Capture instance) carries none of its own — else the referenced image's; the first Softcopy VOI LUT Sequence (0070,0100 group) item applicable to that reference (matched by Referenced Image Sequence, or applying to every reference when a VOI item carries none), else the referenced image's own Window Center/Width, else the full data range; Presentation LUT Shape (0070,0100 group) polarity, defaulting to the referenced image's own Photometric Interpretation when absent; a Displayed Area Selection Sequence crop, at native pixel resolution (every Presentation Size Mode renders the same way — there is no live viewport to "scale to fit" into); and the Graphic Annotation Sequence's Text Objects and Graphic Objects (CIRCLE, ELLIPSE — as an axis-aligned bounding box, not a rotated ellipse — POLYLINE, INTERPOLATED, POINT), layered by Graphic Layer Order, DISPLAY-unit coordinates unsupported (no live viewport to size them against). A rendered page has no window/level control of its own (the window is already baked in) and does not cine-play, matching a Structured Report/ PDF/Waveform document run.

- **Selectable Window Center/Width presets:** any grayscale run whose Window Center (0028,1050) / Window Width (0028,1051) carry more than one value pair — e.g. CT's SOFT TISSUE/BONE/LUNG presets — offers a Preset picker (info column) that applies a chosen pair as the active window; labels come positionally from Window Center & Width Explanation (0028,1055) when present, else "Preset N". A pair repeated verbatim within the instance collapses to one entry, keeping its first label — a run left with a single distinct pair offers no picker rather than a choice between identical windows. The default window shown on first display is still the first pair, per the VOI resolution order above; selecting a preset does not persist across slices of a grouped series run (§ above) — stepping to another slice keeps the last window/level the user set, matching how a real DICOM viewer's W/L control behaves when paging a stack.
- **Calibrated distance measurement (Caliper Tool):** an application feature (View ▸ Caliper Tool) that reports a user-drawn two-point distance in millimeters, resolved per run at index time in priority order: (1) the first item of the Sequence of Ultrasound Regions (0018,6011)'s Physical Delta X (0018,602C) when its Physical Units X Direction (0018,6024) is cm; (2) Pixel Spacing (0028,0030)'s column value; (3) for XA/RF only, Imager Pixel Spacing (0018,1164) divided by the geometric magnification — Estimated Radiographic Magnification Factor (0018,1114), else Distance Source to Detector (0018,1110) ÷ Distance Source to Patient (0018,1111) — an isocenter-plane approximation shown as an **approximate** reading ("~N mm") until the user calibrates on a visible reference (below). No calibration is written back to the data set, and no DICOM measurement/annotation object (e.g. an SR or GSPS) is created — the result is display-only, shown as an on-screen overlay and optionally recorded in a Quiz Mode finding (§4.7). As an application feature beyond the data set, View ▸ Calibrate Caliper… lets the user re-derive the scale by measuring across a reference of known size (a catheter shaft, by French size, or a custom length) — this manual calibration always takes priority over the automatic scale above, persists per run for the session, and is the only path to a scale for a run whose data set has none of the attributes listed above.
- **Not applied / not displayed:** Color, Pseudo-Color, Blending, and Volumetric (MPR / Volume Rendering) Presentation States. Grayscale Softcopy Presentation States (and the XA/XRF

sibling) *are* applied — see the rendering described in §4.4.

- **Rendering:** frames are rendered to 8-bit for display at the instance's **native pixel resolution and aspect ratio** — every modality (X-ray angiography, ultrasound, CT, …) displays through the same freely resizable viewport with no per-modality display mode, no auto-fit, and no letterboxing; a frame larger than the viewport is clipped and can be roamed. The user can additionally fit the whole frame to the window, or zoom from 0.1x to 8x (mouse marquee, ⌘-scroll, trackpad pinch, or the View menu), none of which is written to the data set or affects exported output (§4.7: export/copy are always the acquired resolution). Thumbnails and interactive zoom selections preserve the source aspect ratio. No PS3.14 (GSDF) display calibration is performed.

4.7 Extensions Beyond Part 10 Reading

- Files lacking File Meta Information (raw data sets) are accepted: the transfer syntax is inferred from the detected byte order / VR encoding and the data set is read best-effort.
- `DICOMDIR` located one directory level below the selected folder is accepted (some media place the file-set in a subfolder).
- Encapsulated PDF documents are rendered in-app via Apple PDFKit.
- Loaded runs can be exported to QuickTime (.mov, H.264) via AVFoundation, and the current frame can be copied to the clipboard; both render what the Review pane's frame outline shows — at 1× the whole frame, zoomed in the magnified part inside it — at the instance's full acquired resolution: the frame's size, or the run's ROI's (§4.6) when one is set — for a frame larger than the pane at 1×, the part the pane shows (rows/columns beyond it are cut, never rescaled). An export option ignores the zoom/pan and renders the whole acquired frame cut only by the ROI. With View ▸ Hide ROI on for a run, Copy instead takes the Review image exactly as displayed — the visible area at its on-screen size, ROI ignored; QuickTime export always applies the ROI. These outputs are **not** DICOM objects and carry no DICOM header; any burned-in identification in the pixel data is preserved as-is.
- **Quiz Mode** (View ▸ Quiz Mode) is a non-DICOM, application-only self-study feature: it withholds the display of a study's own SR/Encapsulated PDF instances from the Select strip until the user reveals them, and lets the user record structured findings (Cath/Echo/ Vascular templates) referencing the Caliper Tool's measurements. Neither toggling it nor recording findings reads any additional DICOM attribute or alters the source data set; the saved findings are written as a plain JSON file (not a DICOM SR or other DICOM object) alongside a de-identified run label and the application version, at a location the user chooses.

5. Support of Character Sets

Text is decoded according to Specific Character Set (0008,0005) with the same rules as pydicom, verified against pydicom's character-set samples and synthetic files for every repertoire below. A sequence item without its own Specific Character Set uses the enclosing data set's.

Specific Character Set	Repertoire

absent, ISO_IR 6	Default (read as ISO 8859-1, as pydicom does)
ISO_IR 100 / 101 / 109 / 110 / 148	Latin-1, Latin-2, Latin-3, Latin-4, Latin-5 (Turkish)
ISO_IR 126 / 127 / 138 / 144	Greek, Arabic, Hebrew, Cyrillic (ISO 8859-7 / -6 / -8 / -5)
ISO_IR 166	Thai (TIS 620)
ISO_IR 13	Japanese JIS X 0201 (katakana and roman)
ISO_IR 192	UTF-8
GB18030, GBK	Chinese (stand-alone; any further values are ignored)
ISO 2022 IR 6 / 13 / 87 / 159 / 100 / 101 / 109 / 110 / 126 / 127 / 138 / 144 / 148 / 166	The repertoires above, plus JIS X 0208 (IR 87) and JIS X 0212 (IR 159), selected by ISO 2022 escape sequences
ISO 2022 IR 149	Korean KS X 1001
ISO 2022 IR 58	Chinese GB 2312

Code extensions follow PS3.5 6.1.2.5: each escape sequence selects its repertoire if that repertoire is declared, and the first value's repertoire is restored after each value, Person Name component group, name-part separator (^), and line delimiter (CR, LF, TAB, FF) in LO, SH, ST, LT, UC and UT text. Common misspellings of the defined terms (e.g. `ISO IR 100`, `ISO-IR 144`) are read as the intended term; an unknown term, or an escape sequence for an undeclared repertoire, falls back to the first value. Bytes that cannot be decoded are shown as U+FFFD (). One intentional difference from pydicom: for ISO 2022 IR 58 the escape sequence is removed before decoding, where pydicom leaves its bytes in the text.

Person Name values are displayed with component delimiters (^) replaced by spaces (empty components dropped); the ideographic and phonetic component groups (= delimited, PS3.5 6.2.1) follow the first group in parentheses, e.g. "Yamada Tarou (山田 太郎, やまだ たろう)".

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6. Security

- The application runs entirely offline; no data leaves the machine except by explicit user action (QuickTime export, clipboard copy).
- Media are read-only to the application; Eject unmounts the volume and purges all decoded frames and demographics from memory.
- Decoded pixel data is held in memory only; no image cache is written to disk. Two transient exceptions: an Encapsulated PDF opened externally is written as a temporary PDF file for the system viewer, and a video-transfer-syntax bitstream is written to a temporary file for the AVFoundation decoder and deleted as soon as decoding ends. Both live in the application's sandbox container.
- The application runs in the macOS **App Sandbox**: it can read and write only files and folders the user selects in an Open or Save panel (or drops on it), has no network entitlement, and cannot reach other applications' data.

- The direct-download build is Developer ID signed and Apple-notarized, and stops launching 90 days after first launch.
- No de-identification is performed on export/copy outputs.

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7. Annexes

7.1 IOD Attribute Usage (informative)

Attributes read for indexing and display include: SOP Class UID, Transfer Syntax UID, Modality, Number of Frames, Rows/Columns, Bits Allocated/Stored, Photometric Interpretation, Pixel Data, the frame-rate attributes listed in 4.6, Window Center/Width, the Frame VOI LUT functional-group macro, Modality LUT Sequence, Rescale Slope/Intercept, VOI LUT Sequence, VOI LUT Function, Pixel Aspect Ratio, Pixel Spacing / Imager Pixel Spacing / Pixel Measures (ratio only, except where noted below), Lossy Image Compression / Method / Ratio, Palette Color LUT attributes, Positioner Primary/Secondary Angle (and the Positioner Position macro), the 60xx Overlay Plane attributes, the Display Shutter attributes, the Mask Subtraction Sequence and Recommended Viewing Mode, Patient Name, Patient ID, Patient Birth Date, Study Date, Study Time, Encapsulated Document, MIME Type of Encapsulated Document, Waveform Sequence and its Number of Waveform Channels/Samples, Sampling Frequency, Waveform Bits Allocated, Waveform Sample Interpretation, Waveform Data, Multiplex Group Label, and the per-channel Channel Definition Sequence (Channel Source Sequence, Channel Sensitivity, Channel Sensitivity Correction Factor, Channel Baseline, Channel Sensitivity Units Sequence, Channel Label), Series Instance UID, Instance Number and Window Center & Width Explanation (for multi-slice series grouping and W/L presets — see §4.6), and (DICOMDIR) Referenced File ID.

For the Caliper Tool's calibrated distance measurement (§4.6), Pixel Spacing and Imager Pixel Spacing are additionally read for their absolute value (not just the ratio), together with Sequence of Ultrasound Regions / Physical Delta X / Physical Units X Direction, Estimated Radiographic Magnification Factor, and Distance Source to Detector / Patient.

7.2 Grayscale Rendering (informative)

Grayscale frames are mapped with the DICOM VOI equations (PS3.3 C.11.2.1.2 LINEAR; C.11.2.1.3 LINEAR_EXACT and SIGMOID) over the effective window, or through the data set's VOI LUT table at the default window, then quantized to 8 bits; MONOCHROME1 output polarity is inverted after the VOI stage. Grayscale thumbnails use the same Modality transform and the run's default VOI (VOI LUT table, else the first Window Center/Width pair, else the frame's value range), so a thumbnail matches the run as it first opens.